

CW3-CO1101 Use of Artificial Intelligence/Machine Learning in Healthcare Applications

Introduction

Illness, disability, greater need of health services (due to longer lifespans) and increasing expenditures are challenges faced by health systems from around the world. Overcoming these challenges is critical to improve healthcare and to achieve universal health coverage in countries that haven't currently accomplished it and machine learning, a subset of Artificial Intelligence (AI), may hold the key to transform healthcare systems as we know it today [1]. Machine learning is already being used in radiation therapy, deep learning algorithms are used in organ and lesion segmentation, the end goal being to segment at-risk organs automatically to plan treatments. These algorithms can segment bladder, breast, brain, lung, kidney etc. An example of such an algorithm is U-net (a neural network) [2]. The impact that machine learning could have in healthcare applications will be explored in this report.

Literature Review

AI and machine learning has increased in healthcare as years pass. This is evident as machine learning can predict patients that are at risk of developing diabetes. This can lead to machines providing a healthier diet to the patient. Machine learning and AI is so relevant in healthcare that it is expected to see significant growth in the coming years. Statista a website stated that it has a forecasted compound annual growth rate of over 18.7% from 2023 to 2030[3]. This shows that there is an upwards trend when it comes to machine learning. Also it is known that by 2025 machine learning and artificial intelligence will be the most sought after skill[4]. This will be prevalent in the Indian Job market.

The use of Artificial Intelligence and Machine Learning has become a key factor in Healthcare Applications. The use of cutting-edge technologies holds great potential for transforming the treatment of many diseases. One major disease it has boosted is cancer. Predictive analysis with AI has helped many of those who are diagnosed with cancer, find it early. This helps save the lives of many as it gives them time to receive treatment, which in turn helps them fight the battle and receive therapy. Artificial Intelligence helps by being able to identify patterns that are invisible to the human eye. A clear example of this is when AI was applied to mammographic images, an x-ray image of the breast, it would excel in detecting subtle patterns and anomalies that were not showing in typical traditional screening methods. Since then research has indicated that the use of artificial intelligence can improve breast cancer detection accuracy by as much as 20 percent.[5]

Furthermore, artificial intelligence has also been able to detect rarer cancers better than the traditional screenings provided before its implementation. According to a study, AI is almost twice as accurate as the current approach at determining how aggressive a rare type of cancer is based on scan results. AI was 82% accurate in identifying details that were invisible to the human eye, compared to 44% accuracy in lab analysis.[6] This study not only shows that AI can perform better than the usual screening but also shows us that it exceeds human analytical capabilities. This striking contrast demonstrates how AI has the potential to revolutionise the treatment of rare cancer types by enabling more accurate and efficient diagnosis and treatment

plans. As artificial intelligence advances, its role in reshaping and influencing the future of rare cancer detection will increase and revolutionise cancer treatment.

[3] [Machine Learning - Worldwide | Statista Market Forecast](#)

[4] [https://emeritus.org/in/learn/ai-skills-to-learn/#:~:text=Artificial%20Intelligence%20and%20Machine%20Learning%20\(ML\),sought-after%20skills%20f](https://emeritus.org/in/learn/ai-skills-to-learn/#:~:text=Artificial%20Intelligence%20and%20Machine%20Learning%20(ML),sought-after%20skills%20f)

[5] [https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10312208/#:~:text=Artificial%20intelligence%20\(AI\)%2C%20which,selection%20based%20on%20deep%20knowledge](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10312208/#:~:text=Artificial%20intelligence%20(AI)%2C%20which,selection%20based%20on%20deep%20knowledge).

[6] <https://www.bbc.co.uk/news/health-67264350>

Security, Ethical and Social Issues

The integration of Artificial Intelligence (AI) and Machine Learning (ML) in healthcare applications brings forth critical security, ethical, and social considerations. Security-wise, the vulnerability of health data to cyber threats is a prominent concern [7]. Implementing robust encryption protocols and secure storage systems becomes imperative to safeguard patient information and prevent unauthorised access.

On the ethical front, the utilisation of AI/ML in healthcare necessitates a careful examination of privacy and consent issues. Patients should be adequately informed about data usage, with autonomy granted for granting or denying access [8]. The potential for biases in algorithms poses ethical challenges, requiring continuous scrutiny and refinement of AI models to mitigate differential treatment and healthcare disparities.

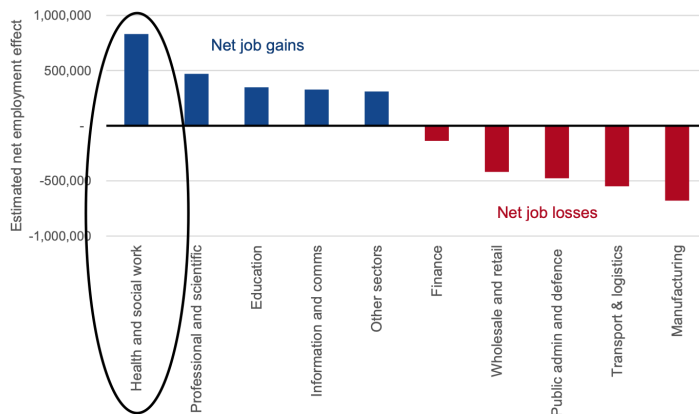
From a social perspective, the adoption of AI/ML in healthcare may reshape employment dynamics. While it creates new opportunities in the tech sector, there's a need to strike a balance to ensure a socially responsible implementation that considers workforce implications [9].

Conclusion

To conclude, the findings highlight the challenges faced by global health systems, including the increasing demand for healthcare due to longer life expectancy, rising costs, and the prevalence of diseases and disabilities. Overcoming these obstacles is essential for achieving universal health coverage, and machine learning, a subset of artificial intelligence (AI), is identified as having the potential to greatly transform healthcare systems. However, as a result, if more and more AI and machine learning are implemented, it raises a question on whether it will be ethical, security issues and other social issues. From an ethical perspective, should something with no emotion take a decision on a patient's life as algorithms look at biases and avoid any treatments that they think are too risky even though that might be a good option for the specific patient's needs. From a social viewpoint, the addition of AI and machine learning may lead to less jobs being available for doctors and nurses and as a result creating unemployment in fields that AI

takes over however *figure 1* shows that the health and social care work environment may see the largest net gain employment increase. For the security concerns, it will have to be made sure that the AI and the machine learning software cannot be affected by any types of cyber-attacks and that the data is secured with robust encryption.

Figure 1 [10]:



The key takeaway is whether implementing this would lead to a negative effect on society due to unemployment already being quite low and other factors such as what job displacement will do to all the people who then must go and work in another industry. Reflecting on the activity, our group was very effective in communicating with each other and completing all the tasks. This was because we all had done another coursework before together and therefore there were not any issues we faced for the completion of this task as such. As a group, we understand all the issues and challenges faced on the subject matter.

As a group, some of the aspects we would like to be changed is to make the task a bit more specific by making the criteria a bit less vague as the more specific it is, the better we can do the task. The information available about AI and machine learning about the healthcare sector was very limited online and therefore it was very difficult to research about this topic as there is not enough up-to-date materials on the internet about the topic we discussed. It was also a bit difficult to balance the conciseness with the inclusion of essential information due to the word limit that we had.

We gained skills such as how to reference using the correct method, working as a team, delegating tasks and overall improved our group's ability to analyse and discuss the topic about using AI and machine learning in healthcare applications.

References

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[9] C. Pazzanese, "Ethical concerns mount as AI takes bigger decision-making role," *Harvard Gazette*, Oct. 26, 2020. <https://news.harvard.edu/gazette/story/2020/10/ethical-concerns-mount-as-ai-takes-bigger-decision-making-role/> , [accessed Dec. 12, 2023]

[10] "The Potential Impact of Artificial Intelligence on UK Employment and the Demand for Skills A report by PwC for the Department for Business, Energy and Industrial Strategy," Aug. 2021. Available: <https://assets.publishing.service.gov.uk/media/615d9a1ad3bf7f55fa92694a/impact-of-ai-on-jobs.pdf> [accessed Dec. 12, 2023]